

# LIZHONG CHEN

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## Education

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- The University of Melbourne** Doctor of Philosophy (PhD) in Statistics 02/2017 – 02/2022  
Supervisors: A/Prof. Guoqi Qian, Prof. Yuriy Kuleshov, Dr Tingjin Chu  
Thesis: Model selection and averaging by Gibbs sampler with a tropical cyclone seasonal forecasting application
- Peking University** Master of Science (M.S.) in Mathematics 09/2013 – 07/2016  
Mentor: A/Prof. Houhong Fan  
Thesis: A Report on Computations of Cohomology Rings and Homotopy Groups of Lie Groups
- Peking University** Bachelor of Science (B.S.) in Mathematics 09/2013 – 07/2016

## Research Experience

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- Smyth Lab, WEHI** *Research Officer* 03/2022 – present  
Supervisor: Prof. Gordon K Smyth

## Publications

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- [1] Jianan Wang, Lizhong Chen, Daniel V Brown, Chris Y Chiu, and Terence P Speed. CMDdemux: an efficient single cell demultiplexing method. *Nucleic Acids Research*, 54(8), 2026.
- [2] Pedro L Baldoni#, Lizhong Chen#, Mengbo Li, Yunshun Chen, and Gordon K Smyth. Dividing out quantification uncertainty enables assessment of differential transcript usage with limma and edgeR. *Nucleic Acids Research*, 53(22), 2025.
- [3] Yunshun Chen, Lizhong Chen, Aaron TL Lun, Pedro L Baldoni, and Gordon K Smyth. edgeR v4: powerful differential analysis of sequencing data with expanded functionality and improved support for small counts and larger datasets. *Nucleic Acids Research*, 53(2), 2025.
- [4] Pedro L Baldoni, Lizhong Chen, and Gordon K Smyth. Faster and more accurate assessment of differential transcript expression with Gibbs sampling and edgeR v4. *NAR Genomics and Bioinformatics*, 6(4), 2024.
- [5] Guoqi Qian#, Lizhong Chen#, and Yuriy Kuleshov. Improving methodology for tropical cyclone seasonal forecasting in the Australian and the South Pacific Ocean regions by selecting and averaging models via Metropolis–Gibbs sampling. *Remote Sensing*, 14(22), 2022.

# contribute equally.

## Preprints

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- [1] Jianan Wang, Lizhong Chen, Rachel Thijssen, Belinda Phipson, and Terence P Speed. GLMsim: a GLM-based single cell RNA-seq simulator incorporating batch and biological effects. *bioRxiv*, 2024.

## Talks and Presentations

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- edgeR quasi-likelihood: correcting deviance for bias** Bioinformatics division seminar, WEHI 08/2022
- Extending edgeR for small counts and large samples** Bioinformatics division seminar, WEHI 05/2023
- Applying edgeR for alternative splicing analysis** Bioinformatics division seminar, WEHI 04/2024
- edgeRv4 with expanded functionality and improved support** Short talk, EuroBioC2025 09/2025
- Analyzing paired count data using edgeR** Bioinformatics division seminar, WEHI 09/2025

## Teaching Experience

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- The University of Melbourne** *Tutor* 07/2017 - 02/2022  
Linear Algebra, Accelerated Mathematics I/II, Complex Analysis, Multivariate Statistical Techniques
- The University of Melbourne** *Assignment and Exam Marker* 07/2018 - 07/2021  
Statistical Modelling, Mathematical statistics
- Peking University** *Teaching assistant* 09/2013 - 07/2016  
Mathematical analysis I/II, Calculus I, Advanced algebra I

## Honors and scholarships

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|-------------------------------------------|-----------------------------|-----------|
| Melbourne Research Scholarship            | The University of Melbourne | 2017-2021 |
| Outstanding Graduate of Beijing           |                             | 2016      |
| Outstanding Graduate of Peking University | Peking University           | 2016      |
| Guanghua Scholarship                      | Peking University           | 2015      |
| National Scholarship                      | Peking University           | 2014      |
| Second-Class Freshmen Scholarship         | Peking University           | 2009-2012 |

## Software Projects

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### IBGS: Iterated Blockwise Gibbs Sampler

IBGS is an MCMC search algorithm to find the best model in the high dimensional data given a model selection criterion such as AIC, BIC and so on. Also, it can identify those most important covariates having the significant influence on the response.

### edgeR: Empirical Analysis of Digital Gene Expression Data in R

edgeR is a Bioconductor R package to perform differential expression analysis for the sequencing data like RNA-seq, ChIP-seq and ATAC-seq. In edgeR v4, we develop the new quasi-likelihood method based on the quadratic mean-variance relationship that we can isolate the biological and technical variation. We improve the quasi-dispersion estimation by the adjusted deviance statistics and support lowly expressed counts data. Besides, we update the computational code from C++ to C for simple structure.

### statmod: Statistical Modeling

statmod is a CRAN R package that is a collection of statistical tools. We introduce a new function `expectedDeviance` to provide an approximation to the expected unit deviance and degree of freedom for Poisson, Binomial, and Negative Binomial distribution, which corrects the bias of chi-square distribution approximation of 1 degree of freedom.

### limma: Linear Models for Microarray and Omics Data

limma is a Bioconductor R package to perform differential expression analysis for the microarray, sequencing and omics data. We update the estimation of the prior distribution in empirical Bayes approach by a two-steps method, combining the local moment estimates and profile likelihood estimates. It improves the performance when the observed degrees of freedom from the models are unequal and relatively small.

## Supervision

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|---------------------------------------------------|---------------------------------------------------------------------|--------|
| <b>Hasini Herath</b>                              | <i>PhD (Medical Biology), University of Melbourne</i>               | 2025 – |
| Co-supervised with Pedro Baldoni and Gordon Smyth |                                                                     |        |
| <b>Xueming Li</b>                                 | <i>PhD (Mathematics and Statistics), University of Melbourne</i>    | 2024 – |
| Co-supervised with Guoqi Qian                     |                                                                     |        |
| <b>Lei Qin</b>                                    | <i>PhD (Medical Biology), University of Melbourne</i>               | 2024 – |
| Co-supervised with Yunshun Chen and Gordon Smyth  |                                                                     |        |
| <b>Shama Deb</b>                                  | <i>Master (Mathematics and Statistics), University of Melbourne</i> | 2025 – |
| Co-supervised with Gordon Smyth                   |                                                                     |        |
| <b>Yuchen Liu</b>                                 | <i>Master (Mathematics and Statistics), University of Melbourne</i> | 2025 – |
| Co-supervised with Tingjin Chu                    |                                                                     |        |

## Referee services

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Applied mathematical modelling, Scientific reports, Nucleic Acids Research, Australian & New Zealand Journal of Statistics